



Universiteit Antwerpen
| Faculteit Wetenschappen

Modelling Of Software Intensive Systems

Assignment 5: DEVS

1st Master computer science
2024-2025

Liam Leirs 20210395
Robbe Teughels 20211127

WaterWay Locks:

Interfaces:

Generator

- Out_ship:
Generates a ship

Queue

- In_ship
Consumes the generated ships of the Generator
- In_update_ship
Update the state of the ship (pop from queue)
- Out_ship_top
Output the Top of queue on update

Load Balancer

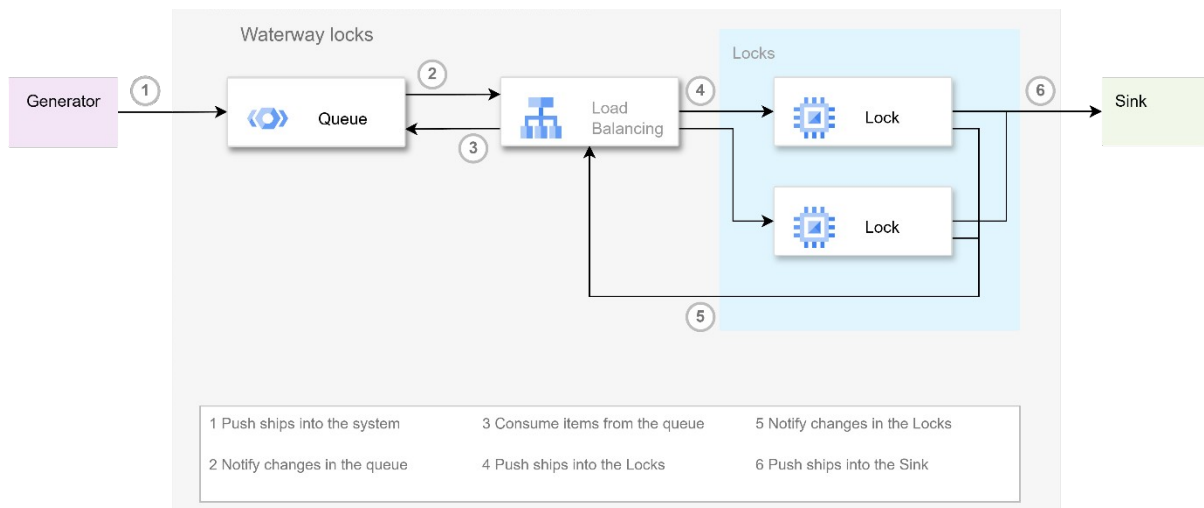
- In_update_queue
Receive updates from the queue
- In_update_lock
Receive updates from the locks
- Out_update_ship
Update the location of a ship
- Out_update_lock[]
Send a ship to a lock

Lock

- In_lock
Receive ships from the loadBalancer
- Out_update_capacity
Update remaining capacity
- Out_sink
Move ships to the sink

Sink

- In_ships
Receive served ships



Protocoles:

1:

The Environment will push ships into the system to be served. These will be captured into a buffer.

2:

When the buffer gets updated, the load balancer will get notified with the new top of the buffer.

4:

With this information combined with the status of all the locks, we can assign a ship to a specific lock by a dedicated gate.

5:

When an update is triggered in a lock, the status is send back to the load balancer. When this happened because of a ship, we also send the ship back.

3:

When a ship is allocated by a lock, we update the buffer to remove this element. (receiving the new top and processing the next ship).

6:

Processed ships are send back to the sink in environment.

Analysis:

With a maximum wait duration of 0, ships experience prolonged queue times as the lock immediately forwards a ship to the sink, allowing only one ship to pass at a time. When longer max waiting durations are used, the time ships spend in the queue does not vary significantly, except in scenarios where smaller ships are prioritized in the round-robin approach. In such cases, the locks tend to fill up with smaller ships, forcing larger ships to wait longer since they can't fit into the locks anymore. The outliers observed in the boxplot correspond to these larger ships, which endure extended wait times compared to their smaller counterparts.

